Generalized model:

$$H = \sum_{k \in BZ} c_k^+ \left[\varepsilon_1 \sigma^x + \varepsilon_2 \left(\cos(ka)\sigma^x + \sin(ka)\sigma^y \right) \right] c_k$$

Can choose $\varepsilon_1 = 1-\alpha$, $\varepsilon_2 = \alpha$ so that

$H(\alpha=0) = H_1$, $H(\alpha=1) = H_2$

Spectrum:

$$\det \begin{pmatrix} -E & \varepsilon_1 + \varepsilon_2 e^{-ika} \\ \varepsilon_1 + \varepsilon_2 e^{ika} & -E \end{pmatrix} = 0 = E^2 - (\varepsilon_1 + \varepsilon_2 e^{-ika})(\varepsilon_1 + \varepsilon_2 e^{ika})$$

$$E^2 = \varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos ka$$

$$E_\pm(k) = \pm\sqrt{\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos ka}$$

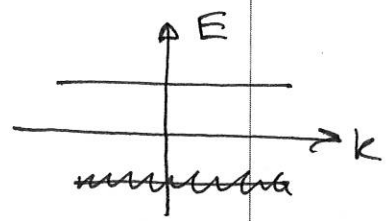
$$\cancel{(1-\alpha)^2 + \alpha^2 + 2(1-\alpha)\alpha \cos ka = 1 - 2\alpha + 2\alpha^2 + (2\alpha - 2\alpha^2)\cos ka}$$

$$\cancel{= 1 - 2\alpha(1-\alpha)(1-\cos ka)}$$

Observe that

for $\varepsilon_2 = 0$, $E_\pm = \pm\varepsilon_1$

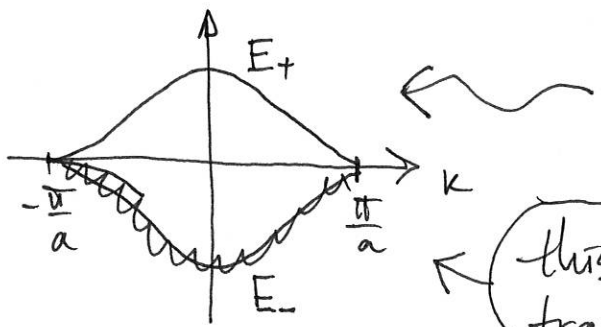
for $\varepsilon_1 = 0$, $E_\pm = \pm\varepsilon_2$



but for $\varepsilon_1 = \varepsilon_2 = \varepsilon$, $E_\pm = \pm\sqrt{2\varepsilon^2 + 2\varepsilon^2 \cos ka} =$

$$= \pm (4\varepsilon^2)^{1/2} \sqrt{\cos^2 \frac{ka}{2}} =$$

$$= \pm 2\varepsilon \left| \cos \frac{ka}{2} \right| \Rightarrow \text{no gap at } k = \pm \frac{\pi}{a}$$



this is phase transition point!

Re-write spectrum slightly

$$E_\pm(k) = \pm\sqrt{\varepsilon_1^2+\varepsilon_2^2-2\varepsilon_1\varepsilon_2+2\varepsilon_1\varepsilon_2\underbrace{(1+\cos ka)}_{2\cos^2\frac{ka}{2}}} =$$

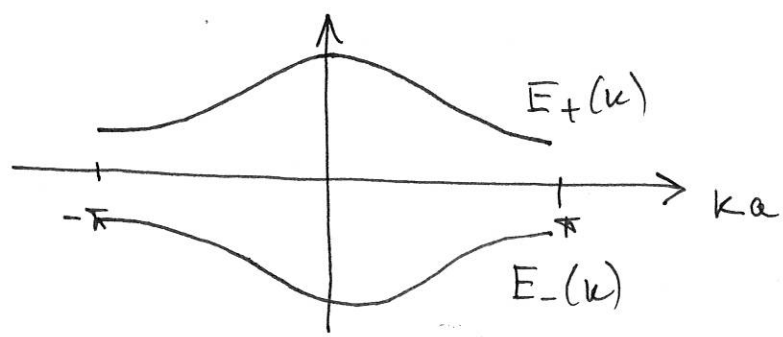
$$= \pm\sqrt{(\varepsilon_1-\varepsilon_2)^2+4\varepsilon_1\varepsilon_2\cos^2\frac{ka}{2}}$$

In this way it is easy to see that the minimum of dispersion is at $ka=\pm\pi$, i.e. at the BZ ends.

Let $k=\frac{\pi}{a}-q$, then $E_\pm(q)\approx\pm\sqrt{(\varepsilon_1-\varepsilon_2)^2+\varepsilon_1\varepsilon_2 q^2a^2}$

$$\left(\cos^2\frac{ka}{2}=\cos^2\left(\frac{\pi}{2}-\frac{qa}{2}\right)\approx\frac{1}{4}(qa)^2\right)$$

The minimal gap between the lower (filled) and upper (empty) bands is $E_+(k=\frac{\pi}{a})-E_-(k=\frac{\pi}{a})=2|\varepsilon_1-\varepsilon_2|$.



$$E_\pm(k)=\pm\sqrt{(\varepsilon_1-\varepsilon_2)^2+4\varepsilon_1\varepsilon_2\cos^2\frac{ka}{2}}$$

$\Rightarrow$ the band structure is identical for the cases of (1) $\varepsilon_1>\varepsilon_2$ and (2) $\varepsilon_1<\varepsilon_2$.

But in fact these are two distinct phases:

$\varepsilon_1>\varepsilon_2$: $W_1=0 \Rightarrow$ "trivial" phase; $\varepsilon_1<\varepsilon_2$: $W_2=+1 \Rightarrow$ "topologically non-trivial"

Topological invariant? Let's see how winding number depends on $\varepsilon_{1,2}$.

$$\hat{n}(k) = \frac{(\varepsilon_1 + \varepsilon_2 \cos ka)\hat{x} + \varepsilon_2 \sin ka\, \hat{y}}{\sqrt{(\varepsilon_1 + \varepsilon_2 \cos ka)^2 + \varepsilon_2^2 \sin^2 ka}} =$$

$$= \frac{(\varepsilon_1 + \varepsilon_2 \cos ka)\hat{x} + \varepsilon_2 \sin ka\, \hat{y}}{\sqrt{\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos ka}}$$

$z = ka$

$$W = \frac{1}{2\pi}\int_{-\pi}^{\pi} dz \left(n_x(z)\frac{dn_y(z)}{dz} - n_y(z)\frac{dn_x(z)}{dz} \right)$$

$$\hat{n}(z) = \frac{(\varepsilon_1 + \varepsilon_2 \cos z)\hat{x} + \varepsilon_2 \sin z\, \hat{y}}{\sqrt{\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos z}}$$

$$W = \frac{1}{2\pi}\int dz \left(\frac{\varepsilon_1 + \varepsilon_2 \cos z}{\sqrt{\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos z}} \frac{d}{dz}\frac{\varepsilon_2 \sin z}{\sqrt{\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos z}} \right.$$

$$\left. - \frac{\varepsilon_2 \sin z}{\sqrt{\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos z}} \frac{d}{dz}\frac{\varepsilon_1 + \varepsilon_2 \cos z}{\sqrt{\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos z}} \right) =$$

$$= \frac{1}{2\pi}\int dz \left\{ \frac{(\varepsilon_1 + \varepsilon_2 \cos z)}{\sqrt{}} \left(\frac{\varepsilon_2 \cos z}{\sqrt{}} - \frac{(\varepsilon_2 \sin z)(-2\varepsilon_1\varepsilon_2 \sin z)}{2(\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos z)^{3/2}} \right) \right.$$

$$\left. - \frac{\varepsilon_2 \sin z}{\sqrt{}} \left(-\frac{\varepsilon_2 \sin z}{\sqrt{}} + \frac{(\varepsilon_1 + \varepsilon_2 \cos z)\,\varepsilon_1\varepsilon_2 \sin z}{(\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1\varepsilon_2 \cos z)^{3/2}} \right) \right\}$$

$$W = \frac{1}{2\pi}\int dz \left\{ \frac{\varepsilon_1\varepsilon_2\cos z + \varepsilon_2^2\cos^2 z + \varepsilon_2^2\sin^2 z}{\varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1\varepsilon_2\cos z} + \right.$$

$$\left. + \frac{\varepsilon_1\varepsilon_2\sin z}{(\varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1\varepsilon_2\cos z)^2}\left[(\varepsilon_1+\varepsilon_2\cos z)\,\varepsilon_2\sin z - \varepsilon_2\sin z\,(\varepsilon_1+\varepsilon_2)\cos z\right]\right\}$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} dz \; \frac{\varepsilon_2^2+\varepsilon_1\varepsilon_2\cos z}{\varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1\varepsilon_2\cos z} =$$

$$= \frac{1}{2\pi}\int dz \; \frac{1}{2}\,\frac{2\varepsilon_1\varepsilon_2\cos z + \varepsilon_1^2+\varepsilon_2^2 - \varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1\varepsilon_2\cos z} =$$

$$= \frac{1}{4\pi}\int_{-\pi}^{\pi} dz \left(1 - \frac{\varepsilon_1^2-\varepsilon_2^2}{\varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1\varepsilon_2\cos z}\right)$$

Thus $W = \frac{1}{4\pi}\,2\pi - \frac{\varepsilon_1^2-\varepsilon_2^2}{4\pi}\,I$

$$W = \frac{1}{2} - \frac{\varepsilon_1^2-\varepsilon_2^2}{4\pi}\,I = \frac{1}{2} - \frac{\varepsilon_1^2-\varepsilon_2^2}{2|\varepsilon_1^2-\varepsilon_2^2|}$$

but $I = \frac{2\pi}{|\varepsilon_1^2-\varepsilon_2^2|}$ (see pp. 129-130)

$$\Rightarrow W = \frac{1}{2}\left(1 - \mathrm{sgn}(\varepsilon_1^2-\varepsilon_2^2)\right)$$

$\Rightarrow$ for $\varepsilon_1 > \varepsilon_2$: $\mathrm{sgn}(\varepsilon_1^2-\varepsilon_2^2) = +1 \Rightarrow W = 0.$

for $\varepsilon_1 < \varepsilon_2$: $\mathrm{sgn}(\quad) = -1 \Rightarrow W = 1$

} topological invariant indeed.

$$\cancel{\frac{1}{4\pi}\int_{-\pi}^{\pi} dz \; \frac{2\varepsilon_2^2+\varepsilon_1\varepsilon_2(e^{iz}+e^{-iz})}{\varepsilon_1^2+\varepsilon_2^2+\varepsilon_1\varepsilon_2(e^{iz}+e^{-iz})}}$$

Denote

$$I = \int_{-\pi}^{\pi} dz \; \frac{1}{\varepsilon_1^2+\varepsilon_2^2+\varepsilon_1\varepsilon_2(e^{iz}+e^{-iz})} = \begin{pmatrix} e^{iz}=x \\ dx = e^{iz}\, i\, dz \\ dz = -i\,\frac{dx}{x} \end{pmatrix} =$$

$$= -i \oint \frac{dx}{x} \; \frac{1}{\varepsilon_1^2+\varepsilon_2^2+\varepsilon_1\varepsilon_2\left(x+\frac{1}{x}\right)} =$$

~~line~~ integral over unit circle

$$= -i \oint \frac{dx}{x} \; \frac{1}{\varepsilon_1\varepsilon_2\, x^2+(\varepsilon_1^2+\varepsilon_2^2)x+\varepsilon_1\varepsilon_2} =$$

$$= \frac{-i}{\varepsilon_1\varepsilon_2} \oint dx \; \frac{1}{x^2+\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2}x+1}$$

Roots $\quad x^2+\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2}x+1=0$

$$\cancel{x = -\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2} \pm \sqrt{\frac{\varepsilon_1^2+\varepsilon_2^2-4\varepsilon_1\varepsilon_2}{\varepsilon_1\varepsilon_2}}}$$

$$x = \frac{1}{2}\left(-\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2} \pm \sqrt{\left(\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2}\right)^2-4}\right) =$$

$$= \frac{1}{2}\left(-\frac{\varepsilon_1^2+\varepsilon^2}{\varepsilon_1\varepsilon_2} \pm \sqrt{\left(\frac{\varepsilon_1^2-\varepsilon_1^2}{\varepsilon_1\varepsilon_2}\right)^2}\right)$$

$$X_{1,2} = \frac{1}{2}\left(-\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2} \pm \frac{|\varepsilon_1^2-\varepsilon_2^2|}{\varepsilon_1\varepsilon_2}\right)$$

$$X_1 \cdot X_2 = \frac{1}{2}\left(-\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2} + \frac{|\varepsilon_1^2-\varepsilon_2^2|}{\varepsilon_1\varepsilon_2}\right)\frac{1}{2}\left(-\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2} - \frac{|\varepsilon_1^2-\varepsilon_2^2|}{\varepsilon_1\varepsilon_2}\right) =$$

$$= \frac{1}{4}\left[\left(\frac{\varepsilon_1^2+\varepsilon_2^2}{\varepsilon_1\varepsilon_2}\right)^2 - \left(\frac{\varepsilon_1^2-\varepsilon_2^2}{\varepsilon_1\varepsilon_2}\right)^2\right] =$$

$$= \frac{1}{4}\left(\frac{2\varepsilon_1^2\varepsilon_2^2}{\varepsilon_1^2\varepsilon_2^2} - \frac{-2\varepsilon_1^2\varepsilon_2^2}{\varepsilon_1^2\varepsilon_2^2}\right) = 1$$

$\Rightarrow$ hence one root is inside the unit circle, and the other is outside.

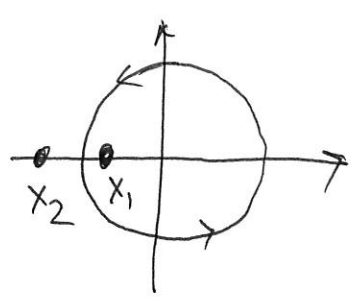
let $\varepsilon_1 > \varepsilon_2$:

$$X_{1,2} = \frac{1}{2\varepsilon_1\varepsilon_2}\left(-\varepsilon_1^2-\varepsilon_2^2 \pm (\varepsilon_1^2-\varepsilon_2^2)\right) = \begin{cases} -\frac{\varepsilon_2}{\varepsilon_1} & (+) \\ -\frac{\varepsilon_1}{\varepsilon_2} & (-) \end{cases}$$

let $\varepsilon_2 > \varepsilon_1$:

$$X_{1,2} = \frac{1}{2\varepsilon_1\varepsilon_2}\left(-\varepsilon_1^2-\varepsilon_2^2 \pm (\varepsilon_2^2-\varepsilon_1^2)\right) = \begin{cases} -\frac{\varepsilon_1}{\varepsilon_2} & (+) \\ -\frac{\varepsilon_2}{\varepsilon_1} & (-) \end{cases}$$

So $|X_1| < 1$ in both cases:

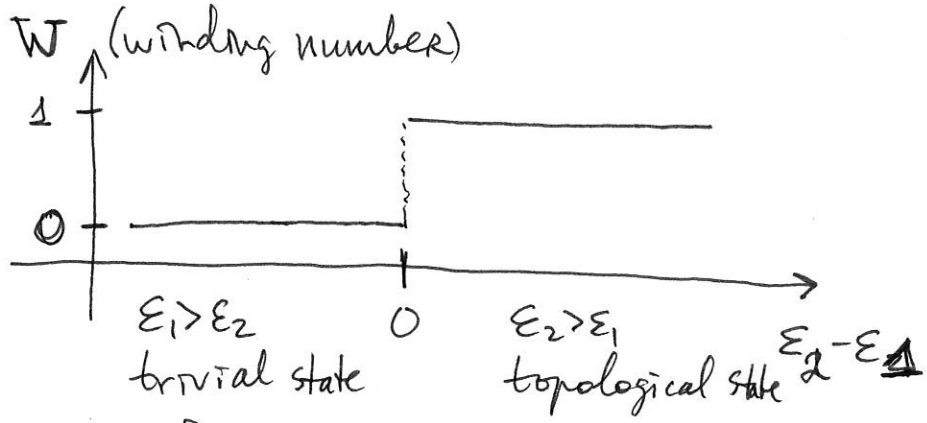


$$I = \frac{-i}{\varepsilon_1\varepsilon_2}\oint \frac{dx}{(x-x_1)(x-x_2)} =$$

$$= \frac{-i}{\varepsilon_1\varepsilon_2}\,\frac{2\pi i}{x_1-x_2} = \frac{2\pi}{\varepsilon_1\varepsilon_2\,\frac{|\varepsilon_1^2-\varepsilon_2^2|}{\varepsilon_1\varepsilon_2}} \; !$$

nice

Winding number changes from $W=0$ (trivial) to $W=1$ (topological phase) at the phase transition point where the <u>gap</u> $2|\varepsilon_1-\varepsilon_2|$ between the $E_\pm(k)$ bands <u>closes</u>.



$$W = \frac{1}{2}\left[1 - \mathrm{sgn}\left(\varepsilon_1^2-\varepsilon_2^2\right)\right]$$

Two states with identical band structure are distinguished by <u>topological invariant</u> W. Being given by the integral over the whole BZ, this invariant is highly <u>non-local</u> object which distinguishes different topological states.

Pretty amazing!